$$\min \quad w^H R w + \lambda_0 \|w\|_1 + \lambda_1 \|A_{SL}^H w\|_1$$

$$\text{s.t.} \quad w^H R_s w \geq 1$$

$$\min \quad f(u) + g(w) \quad \text{s.t.} \quad u = w$$

$$f(u) = u^H R u + I_{\{u^H R_s u \geq 1\}}(u)$$

$$g(w) = \lambda_0 H(w) + \lambda_1 H(A_{SL}^H w)$$

$$H(w) = \|w\|_1 = \sum_{m=1}^{M} h(w_m)$$

$$h(w) = |w|$$

g Lipschitz Continous

$$L_h = 1, \quad L_H = \sqrt{M}$$

Pf: $H(w) = \|w\|_1$

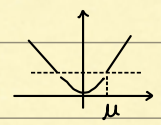
$$\big| \|w\|_1 - \|v\|_1 \big| \leq \|w - v\|_1 \leq \sqrt{M} \|w - v\|_2$$

$$L_g = \lambda_0 \sqrt{M} + \lambda_1 \sqrt{N} \|A_{SL}\|_2$$

Pf: $g(w) = \lambda_0 H(w) + \lambda_1 H(A_{SL}^H w)$

$$\begin{aligned}
|g(w) - g(v)| &= \big| \lambda_0 (H(w) - H(v)) \\
&\quad + \lambda_1 (H(A_{SL}^H w) - H(A_{SL}^H v)) \big| \\
&\leq \lambda_0 |H(w) - H(v)| + \lambda_1 |H(A_{SL}^H w) - H(A_{SL}^H v)| \\
&\leq \lambda_0 \sqrt{M} \|w - v\|_2 + \lambda_1 \sqrt{N_{SL}} \|A_{SL}^H (w - v)\|_2 \\
&\leq (\lambda_0 \sqrt{M} + \lambda_1 \sqrt{N_{SL}} \|A_{SL}\|_2) \|w - v\|_2
\end{aligned}$$

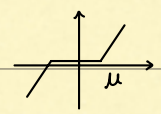
Moreau Envelope



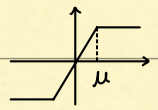
$$h(w) = |w|$$

$$h_\mu(w) := \inf_v \frac{1}{2\mu} |w - v|^2 + h(w) = \begin{cases} |w| - \frac{\mu}{2}, & |w| \geq \mu \\ \frac{|w|^2}{2\mu}, & |w| < \mu \end{cases}$$

$$\text{prox}_h^\mu(w) = \arg\min_v \frac{1}{2\mu} |w - v|^2 + h(w) = S_\mu(w)$$



$$\nabla h_\mu(w) = \frac{\partial}{\partial \bar{w}} h_\mu(w) = \begin{cases} \text{sgn}(w), & |w| \geq \mu \end{cases}$$



$\left| \frac{w}{\mu} \right.$, $|w| < \mu$

$L_{\nabla h_\mu} = \frac{1}{\mu}$

$H(w) = \|w\|_1$ Separable

$H_\mu(w) := \inf_v \frac{1}{2\mu}\|w-v\|^2 + H(w) = \sum_{m=1}^{M} h_\mu(w_m)$

$\mathrm{prox}_H^\mu(w)_{M\times 1} = [\, h_\mu(w_m)\,]_{M\times 1}$

$\nabla H_\mu(w)_{M\times 1} = [\nabla h_\mu(w_m)]_{M\times 1}$

$L_{\nabla H_\mu} = \frac{1}{\mu}$

Pf: Lemma 3.1 in "Variable Smoothing ..."

$\nabla H_\mu(w) = \frac{1}{\mu}(w - \mathrm{prox}_g^\mu(w))$

$\|\nabla H_\mu(w) - \nabla H_\mu(v)\| = \frac{1}{\mu}\|w-v-(\mathrm{prox}_g^\mu(w) - \mathrm{prox}_g^\mu(v))\|$

由于 g 凸, $\langle w-v, \mathrm{prox}_g^\mu(w) - \mathrm{prox}_g^\mu(v)\rangle$

$\geq \|\mathrm{prox}_g^\mu(w) - \mathrm{prox}_g^\mu(v)\|^2$

故 $\|w-v-(\mathrm{prox}_g^\mu(w) - \mathrm{prox}_g^\mu(v))\|^2$

$\leq \|w-v\|^2 - \|\mathrm{prox}_g^\mu(w) - \mathrm{prox}_g^\mu(v)\|^2 \leq \|w-v\|^2$

即 $\|\nabla H_\mu(w) - \nabla H_\mu(v)\| \leq \frac{1}{\mu}\|w-v\|$

$g_\mu(w) := \lambda_0 H_\mu(w) + \lambda_1 H_\mu(A_{SL}^H w)$

此处 g_μ 不是 g 的 envelope

$\nabla g_\mu(w) = \lambda_0 \nabla H_\mu(w) + \lambda_1 A_{SL} \nabla H_\mu(A_{SL}^H w)$

$L_{\nabla g_\mu} = (\lambda_0 + \lambda_1 \|A_{SL}\|_2^2)\frac{1}{\mu}$

SDD-ADMM (Scaled Dual Descent)

$\min f(u) + g(w)$ s.t. $u = w$

$\mathcal{L}(u, w, z; \mu)$

$= f(u) + g(w) + \mathrm{Re}\langle u-w, z\rangle + \frac{\rho}{2}\|u-w\|_2^2$

$= f(u) + g(w) + \frac{\rho}{2}\|u-w+\frac{1}{\rho}z\|^2 - \frac{1}{2\rho}\|z\|^2$

$$\mu^k = \frac{1}{k^{1/3}}, \quad L^k := L_{\nabla g_{\mu^k}} = (\lambda_0 + \lambda_1 \|A_{SL}\|_2^2)\, k^{1/3}$$

$$u^{k+1} = \operatorname{argmin} f(u) + \frac{\rho}{2}\|u - w^k + \frac{1}{\rho} z^k\|^2 + \frac{\alpha}{2}\|u - u^k\|^2$$

$$w^{k+1} = \operatorname{argmin} \langle \nabla g_{\mu^k}(w), \; w - w^k \rangle + \frac{\rho}{2}\|u^{k+1} - w + \frac{1}{\rho} z^k\|^2 + \frac{\gamma L^k}{2}\|w - w^k\|^2$$

$$z^{k+1} = \frac{\tau}{1+\tau}\left[\lambda^k - \frac{\rho}{\omega\tau}(u^{k+1} - w^{k+1})\right]$$

v^k decreasing ↘

$$\mathcal{L}(u^{k+1} \cdot\cdot) - \mathcal{L}(u^k \cdot\cdot) \leq -\frac{\alpha}{2}\|\Delta u^{k+1}\|^2$$

$$\mathcal{L}(w^{k+1} \cdot\cdot) - \mathcal{L}(w^k \cdot\cdot) \leq -\frac{\gamma - 1}{2} L^k \|\Delta w^{k+1}\|^2$$

$$\operatorname{Re}\langle \nabla g_{\mu^k}(z^k), \; \Delta z^{k+1} \rangle + \frac{\beta^k}{2}\|\Delta z^{k+1}\|^2 + \frac{\rho}{2}\|x^{k+1} - z^{k+1} + \frac{1}{\rho}\lambda^k\|^2$$

$$\leq \frac{\rho}{2}\|x^{k+1} - z^k + \frac{1}{\rho}\lambda^k\|^2$$

$$g_{\mu^k}(z^{k+1}) \leq g_{\mu^k}(z^k) + \operatorname{Re}\langle \nabla g_{\mu^k}(z^k), \; \Delta z^{k+1} \rangle + \frac{L^k}{2}\|\Delta z^{k+1}\|^2$$

$$\mathcal{L}(z^{k+1} \cdot\cdot) - \mathcal{L}(z^k \cdot\cdot) = \operatorname{Re}\langle u^{k+1} - w^{k+1}, \; \Delta z^{k+1} \rangle$$

$$u^{k+1} - w^{k+1} = \frac{\omega\tau}{\rho}\left(z^k - \frac{1+\tau}{\tau} z^{k+1}\right)$$

$$= -\frac{\omega\tau}{\rho}\|\Delta z^{k+1}\|^2 - \frac{\omega}{\rho}\operatorname{Re}\langle z^{k+1}, \; \Delta z^{k+1} \rangle$$

$$\mathcal{L}(\cdot\cdot \mu^{k+1}) - \mathcal{L}(\cdot\cdot \mu^k) \leq g_{\mu^{k+1}}(w^{k+1}) - g_{\mu^k}(w^{k+1})$$

$$H_{\mu_2}(w) \leq H_{\mu_1}(w) + \frac{1}{2}\frac{\mu_1}{\mu_2}(\mu_1 - \mu_2) L_H^2, \quad \forall w$$

$$g_{\mu_2}(w) = \lambda_0 H_{\mu_2}(w) + \lambda_1 H_\mu(A_{SL}^H w)$$

$$\leq \lambda_0\left[H_{\mu_1}(w) + \frac{1}{2}\frac{\mu_1}{\mu_2}(\mu_1 - \mu_2) M\right] + \lambda_1\left[H_{\mu_1}(A_{SL}^H w) + \frac{1}{2}\frac{\mu_1}{\mu_2}(\mu_1 - \mu_2) N_{SL}\right]$$

$$= g_{\mu_1}(w) + \frac{1}{2}\frac{\mu_1}{\mu_2}(\mu_1 - \mu_2)(\lambda_0 M + \lambda_1 N_{SL})$$

$$\leq \frac{1}{2}\frac{\mu^k}{\mu^{k+1}}(\mu^k - \mu^{k+1})(\lambda_0 M + \lambda_1 N_{SL})$$

$$\leq 2^{-2/3}(\mu^k - \mu^{k+1})(\lambda_0 M + \lambda_1 N_{SL}) \qquad \frac{\mu^k}{\mu^{k+1}} = \left(\frac{k+1}{k}\right)^{1/3} \leq 2^{1/3}$$

$$\nu^k := \mathcal{L}(u^k, w^k, z^k; \mu^k) + \frac{\omega}{2\rho}\|z^{k+1}\|^2 + 2^{-2/3}\mu^k(\lambda_0 M + \lambda_1 N_{SL})$$

$$\nu^{k+1} - \nu^k \leq -\frac{\alpha}{2}\|\Delta u^{k+1}\|^2 - \frac{\gamma-1}{2}L^k\|\Delta w^{k+1}\|^2$$

$$-\frac{\omega\tau}{\rho}\|\Delta z^{k+1}\|^2 - \frac{\omega}{\rho}\mathrm{Re}\langle z^{k+1}, z^{k+1} - z^k\rangle$$

$$+\frac{\omega}{2\rho}\|z^{k+1}\|^2 - \frac{\omega}{2\rho}\|z^k\|^2$$

$$= -\frac{\alpha}{2}\|\Delta u^{k+1}\|^2 - \frac{\gamma-1}{2}L^k\|\Delta w^{k+1}\|^2$$

$$-\frac{\omega}{\rho}\left(\tau + \frac{1}{2}\right)\|\Delta z^{k+1}\|^2$$

ν^k lower bounded, converge $\nu^{k+1} \longrightarrow \tilde{\nu}$

$\nu^k \geq 0$ If $\omega > 1$

λ^k Bounded / Residual Bounded

$z' = 0$, $\mu' = 1$

$$\nu^k \leq \nu' = f(u') + g(w') + \frac{\rho}{2}\|u' - w'\|^2 + 2^{-2/3}(\lambda_0 M + \lambda_1 N_{SL})$$

$$\nu^k = f(u^k) + g(w^k) + \mathrm{Re}\langle u^k - w^k, z^k\rangle + \frac{\rho}{2}\|u^k - w^k\|^2$$

$$+\frac{\omega}{2\rho}\|z^k\|^2 + 2^{q-1}\mu^k\sum_{i=1}^{p} L_i^2$$

$$\frac{\rho}{4}\|u^k - w^k\|^2 + 2\mathrm{Re}\left\langle\frac{\sqrt{\rho}}{2}(u^k - w^k), \frac{1}{\sqrt{\rho}}z^k\right\rangle + \frac{1}{\rho}\|z^k\|^2 \geq 0$$

$$\mathrm{Re}\langle u^k - w^k, z^k\rangle \geq -\frac{\rho}{4}\|u^k - w^k\|^2 - \frac{1}{\rho}\|z^k\|^2$$

$$\geq (f+g)_{\min} + \frac{\rho}{4}\|u^k - w^k\|^2 + \frac{1}{\rho}\left(\frac{\omega}{2} - 1\right)\|z^k\|^2$$

$$\geq \frac{\omega - 2}{2\rho}\|z^k\|^2$$

$\|z^k\|^2 \leq \frac{2\rho}{\omega - 2}\nu'$ If $\omega > 2$

$\|u^k - w^k\|^2 \leq \frac{4}{\rho}\nu'$

ε - Stationary point

(u, w) is ε - Stationary point

$\exists \check{z}$, $\text{dist}(-\check{z}, \partial f(u)) < \varepsilon$ and $\| u - w \| < \varepsilon$

$\min f(u) + g_\mu(w)$ s.t. $u = w$

Lagrangian $f(u) + g_\mu(w) + \text{Re}\langle u - w, z \rangle$

∂_u : $0 \in \partial f(u) + \check{z}$ For some unknown $\check{z}$

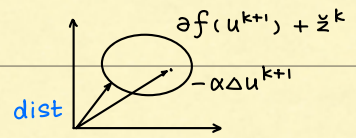
∇_w : $0 =$

Dual Residual Bounded

$$u^{k+1} = \text{argmin}\ f(u) + \frac{\rho}{2} \| u - w^k + \frac{1}{\rho} z^k \|^2 + \frac{\alpha}{2} \| u - u^k \|^2$$

$$0 \in \partial f(u^{k+1}) + \underbrace{\rho (u^{k+1} - w^k + \frac{1}{\rho} z^k)}_{\check{z}^k} + \alpha \Delta u^{k+1}$$

$$-\alpha \Delta u^{k+1} \in \partial f(u^{k+1}) + \check{z}^k$$



$$\text{dist}(-\check{z}^k, \partial f(u^{k+1})) = \text{dist}(0, \partial f(u^{k+1}) + \check{z}^k)$$

$$< \alpha \| \Delta u^{k+1} \|^2$$

$$w^{k+1} = \text{argmin} \langle \nabla g_{\mu^k}(w^k), w - w^k \rangle + \frac{\rho}{2} \| u^{k+1} - w + \frac{1}{\rho} z^k \|^2 + \frac{\gamma L^k}{2} \| w - w^k \|^2$$